Axles and alignment

antocciphysics | TEACHER KEY | CPA

Learning target: Use measurements and observations to improve straight rolling.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 18; 5 course days later.*
35 J; heating or sound are possible.

02. Explain a worked example (worked)
Parallel axle lines, perpendicular intended travel, positive wheel clearance; actual dimensions recorded when measured.

03. Look and explain (model)
Student marks measurable alignment and free clearance.

04. Build the solution (practice)
Forward distance 100 cm; lateral offset +8 cm right. Use repeated actual measurements.

05. Use the idea (reason)
No; wheel rubbing, unequal wheel sizes, surface slope or push could matter. Inspect and repeat controlled trials.

06. Choose a method (mixed) *Review: Unit 7, Day 11; 12 course days later.*
E=Pt=24 J.

07. Exit check (exit)
Specific recorded change with a measurable outcome such as reduced offset or freer rolling.

08. STRETCH: Apply it another way (stretch)
Possibilities unequal wheel drag, skewed axles, unequal effective wheel diameters; unpowered roll/spin checks and one-variable changes distinguish them.

Teaching note: Use actual observations for the build record. Supplied numbers are separate practice examples.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 18; 5 course days later.*
$|emf| = N|\Delta\Phi|/\Delta t = 0.7$ V.

02. Explain a worked example (worked)
Parallel axle lines, perpendicular intended travel, positive wheel clearance; actual dimensions recorded when measured.

03. Represent and interpret (model)
Student marks measurable alignment and free clearance.

04. Solve and justify (practice)
3.43° to the side; magnitude 2.004 m.

05. Analyze the reasoning (reason)
No; wheel rubbing, unequal wheel sizes, surface slope or push could matter. Inspect and repeat controlled trials.

06. Choose a method (mixed) *Review: Unit 7, Day 11; 12 course days later.*
$V = 24$ V; $P = IV = I^2R = 144$ W.

07. Exit check (exit)
Repeatability and variation matter; a single run may reflect noise or different conditions.

08. STRETCH: Apply it another way (stretch)
Mean 0.22 m left; range 0.20-0.24 m. Change one alignment variable; keep track/release and trial count fixed, retain all results.

Teaching note: Use actual observations for the build record. Supplied numbers are separate practice examples.